

MTH165: Mathematics for Engineers

100-Percentile Master Syllabus Blueprint, Curated Video Hub & Exam Strategy

Author: Saikat Koner (Black Hat Coders 108) | Target: Grade 'O' (100/100) | Credits: 4 (L:3 T:1 P:0)

1. Course Units & High-Yield Mathematical Invariants

Unit / Code	Subject Area & Core Mathematical Topics	Standard Reference & Exam Scoring Triggers
Unit I Matrices	Rank via Row Echelon & Normal forms; System consistency $AX=B$ (Rouché-Capelli); Eigenvalues & Eigenvectors ($ A - \lambda I =0$); Cayley-Hamilton Theorem $P(A)=0$, A^{-1} calculation.	B.S. Grewal Ch 2 / Jain & Iyengar Ch 2 - Verify: $\sum \lambda_i = \text{tr}(A)$, $\prod \lambda_i = \det(A)$ - Matrix inverse via CHT: $A^{-1} = (1/c(A))(A^2 - c(A)A + c(A)I)$.
Unit II Diff. Calc.	Parametric (d^2y/dx^2), Implicit & Logarithmic differentiation; Leibniz's n-th derivative rule; Rolle's, LMVT, CMVT; Taylor's & Maclaurin's series; L'Hôpital across 7 forms; Extrema.	B.S. Grewal Ch 4 & 5 / NCERT XII - Leibniz: $(uv)^{(n)} = \sum C_n^r u^{(r)} v^{(n-r)}$ 1^∞ to L'Hôpital: $\ln y = g(x) \ln f(x) \rightarrow e^{\lim}$.
Unit III Integrals	Integration by substitution & parts (ILATE / Bernoulli); Partial fractions; King's property $\int f(a+b-x)dx$; Even/Odd & Split properties; Wallis & Gamma reduction formulas.	NCERT XII Part II / Grewal Ch 6 Wallis: $\int_0^{\pi/2} \sin^n x \cos^n x \, dx = \frac{\Gamma((m+1)/2)\Gamma((n+1)/2)}{2\Gamma((m+n+2)/2)}$.
Unit IV Multivariate	Limits (Path test $y=mx$), Total differential $df = f_x dx + f_y dy$; Euler's theorem on homogeneous functions; Extrema of $f(x,y)$ ($r_t - s^2$); Lagrange's undetermined multipliers.	Jain & Iyengar Ch 5 / Grewal Ch 5 - Euler: $x u_x + y u_y = n F(u)/F'(u)$ - Extrema: $r_t - s^2 > 0, r > 0$ (Min); $r < 0$ (Max).
Unit V Multiple Int.	Double integrals (change of order by switching vertical to horizontal strips); Triple integrals; Jacobians in Polar ($r \, dr \, d\theta$), Cylindrical & Spherical; Area ($\iint dx \, dy$) & Volume ($\iiint dx \, dy \, dz$).	B.S. Grewal Ch 7 / Jain & Iyengar Ch 5 - Sketch region R and mark vertices. - Polar: $dx \, dy = r \, dr \, d\theta$; Spherical: $\rho^2 \sin \phi \, d\rho \, d\phi \, d\theta$.
Unit VI Fourier	Dirichlet's conditions; Euler's Fourier formulas (a_n, a_m, b_n) over $[-L, L]$; Even ($b_n=0$) & Odd ($a_n=a_m=0$) series; Half-range Sine/Cosine expansions; Parseval's identity & series deductions.	Jain & Iyengar Ch 10 / Grewal Ch 10 - Discontinuity value: $[f(x_n)] + f(x_{n+1})/2$ - Parseval: $(1/L) \int_{-L}^L [f(x)]^2 \, dx = a_0^2/2 + \sum (a_n^2 + b_n^2)$.

2. Curated Video Channels & 4-Tier Scoring Strategy

Topic / Unit	Handpicked Video Channel & Key Concepts	100-Percentile Defense Answer
Unit I (Matrices)	Dr. Gajendra Purohit : Rank (Echelon & Normal form), $AX=B$ consistency, CHT inverse.	Write general characteristic equation and check eigenvalue trace/det invariants before finding eigenvectors.
Unit II (Diff Calc)	BSV Maths : Leibniz rule for $(uv)^{(n)}$ proofs, Rolle's/LMVT/Taylor series expansions.	Differentiate base equation twice, clear fractions/roots, then apply Leibniz rule term by term.
Unit III (Integrals)	Dr. Gajendra Purohit : King's property $\int f(a+b-x)dx$ and Wallis Gamma reductions.	Add I and modified I equations to eliminate denominator terms before evaluating.
Unit IV (Multivariate)	Dr. Gajendra Purohit : Euler's theorem on homogeneous functions and $r_t - s^2$ extrema.	Apply composite Euler formula $x u_x + y u_y = n F(u)/F'(u)$ and construct 2nd derivative discriminant table.
Unit V (Multiple Int)	BSV Maths : Change of Order strip switching, Polar/Spherical Jacobians.	Draw 2D Cartesian sketch of region R, shade bounded area, and flip from vertical to horizontal strip.
Unit VI (Fourier)	Dr. Gajendra Purohit : Dirichlet conditions, Half-range cosine/sine, Parseval deductions.	Check even/odd symmetry first to eliminate a_n, a_m , or b_n , and state jump discontinuity convergence.